Assylkhan Geniyat

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Research Interests: Large Language Models, Retrieval-Augmented Generation, Agentic AI, Reasoning, Efficient AI Systems

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Education

Boston University, Graduate School of Arts & Sciences — Boston, MA, USA
Master of Science in Artificial Intelligence — Jan. 2026 – May 2027 (Expected)

- **GPA: 4.00 / 4.00**

Astana IT University — Astana, Kazakhstan
Bachelor of Science in Computer Science — Sep. 2022 – June 2025

- **GPA: 3.65 / 4.00**; graduated with Honors; **ranked 2nd of 120** in the Computer Science major.
- Bachelor's thesis: *AI for Drug Discovery Using Molecular Docking.*

Research Experience

First-Author Research – Retrieval-Augmented Generation — 2026
Boston University, Graduate School of Arts & Sciences
Research mentor: Prof. Aaron Mueller, Department of Computer Science

- Led a first-author study asking whether training a RAG reranker on answer-bearing passages (**literal-match labels**) yields better generated answers than document-level labels.
- Designed controlled experiments with **frozen protocols** and a prespecified success criterion: **Qwen3-Reranker-0.6B with LoRA**, hybrid BM25/dense retrieval, 4 chunking strategies, fixed **Qwen3.5-9B** generator.
- Evaluated in-domain (**208 held-out TechQA questions**) and under transfer (**273 Doc2Dial dialogue turns**, 3 paired seeds), with bootstrap and paired sign-flip uncertainty analysis.
- **Key finding:** clear gains in retrieval quality (NDCG@4) barely changed answer F1, so retrieval metrics alone are an unreliable guide for reranker supervision.
- Led conceptualization, methodology, software, analysis and writing.

Research Assistant — Dec. 2023 – Jul. 2024
Astana IT University
Supervisor: Ida Akhmetvaliyeva

- Conducted research on computational methods for analyzing and generating effective Russian-language media headlines, combining content analysis with reader surveys and interviews.
- Co-developed a **Telegram bot** that suggests and explains more engaging headlines (TensorFlow, pandas, NumPy), using a dataset of **1M+ news headlines**.
- Studied collaborative filtering and recommender systems (cosine and Tanimoto similarity, k-nearest neighbors) for personalizing headline suggestions.
- Contributed to experimental design, data analysis, evaluation and writing of a peer-reviewed IEEE paper; presented the work and received **2nd place among 27 AI research papers**.

Publications & Manuscripts

1. **A. Geniyat**, Z. Fakhriddinov, A. Gulati. "Does Literal-Match Reranker Supervision Improve Generated Answers in RAG? An In-Domain and Transfer Study." *Manuscript under review*, Engineering Applications of Artificial Intelligence (EAAI), 2026.
2. Z. Fakhriddinov, **A. Geniyat**, A. Gulati. "Answer Support Is Not Citation Correctness: A Matched-Evidence Study of Local Scientific RAG." *Manuscript under review*, Engineering Applications of Artificial Intelligence (EAAI), 2026.
3. D. Alkhaidar, **A. Geniyat**, I. Akhmetvaliyeva. "The Role of Computer Technologies in Formulating Effective Headlines for Russian-Language Media." In *2024 IEEE AITU: Digital Generation*, Astana, Kazakhstan, 2024, pp. 140–145. DOI

Professional Experience

Lecturer, AI & Data Science (Part-time) — Sep. 2026 – Present
Astana Hub, Tech Orda Program

- Designed the AI and data science curriculum and training program for the course.
- Teach AI, data science and machine learning to a cohort of **40 students**.

Instructional Technology Developer (Data & Automation) — Jan. 2026 – Present
Boston University — Boston, MA

- Develop AI agents and automation workflows supporting instructional technology, university operations, and data-driven decision making.
- Contributed to an **AI-powered faculty analytics platform** designed to support teaching evaluation, feedback analysis, and instructional improvement at university scale, with potential applicability across Boston University's **4,500+ faculty members**.
- Build API and webhook integrations connecting learning management systems, analytics platforms, databases, and internal university tools.
- Develop automated data pipelines and reporting workflows supporting faculty and administrative decision making.

Machine Learning Engineer (Primary AI/ML Engineer) Jul. 2025 – Dec. 2025
National Bank of Kazakhstan Astana, Kazakhstan

- Served as the **primary AI/ML engineer** responsible for end-to-end design, development, deployment, and technical ownership of **five production AI/ML systems**.
- Built and deployed an enterprise AI-agent and retrieval system serving **6+ internal departments**, enabling semantic search and document question answering across more than **80,000 financial and legal documents**.
- Developed and deployed an anti-fraud ML pipeline for cross-border transaction analysis, identifying suspicious transaction chains exceeding **$500K and achieving 89% recall** through geographic and beneficiary-based pattern analysis.
- Designed the end-to-end RAG architecture, including retrieval, embeddings, metadata-aware search, reranking, vector search, and LLM-based answer generation.

Machine Learning Engineer Mar. 2025 – May 2025
Ministry of Energy of Kazakhstan, Situational-Analytical Center Astana, Kazakhstan

- Developed and deployed a multilingual NLP classification system processing approximately **25,000 news articles per day** in three languages, achieving **97.2% classification accuracy**.
- Built and curated a domain-specific dataset of approximately **100,000 news articles** for a government ministry in Kazakhstan, including data collection, cleaning, and label preparation for model training and evaluation.
- Built and deployed containerized NLP inference services and APIs, integrating trained models into production data pipelines and government-facing systems.
- Developed classification systems for **five government ministries**, improving average recall from **78% to 90%**.

Honors, Scholarships & Selected Training

- **Bolashak Presidential Scholarship** (International Scholarship of the President of the Republic of Kazakhstan) – fully funded government scholarship supporting graduate study at Boston University.
- **Boston University Merit Scholarship** – $20,000.
- **2nd Place, AI Research Paper Competition** – selected among 27 AI research papers, 2024.
- **Full-Tuition Merit Scholarship**, Astana IT University.
- **Samsung Innovation Campus Microqualification** – GPA: **3.91 / 4.00**; final project evaluated as **Excellent**.

Academic Leadership & Service

Conference Organizer, 2024 IEEE AITU: Digital Generation [IEEE] Sep. 2023 – Apr. 2024
Astana IT University

- Served on the organizing team of the **first IEEE student conference held in Kazakhstan**, leading IT operations, developing the official conference website, and providing technical support.
- Helped deliver an international conference that received **200 paper submissions from 7 countries**, with **83 papers** accepted into the IEEE proceedings.

Vice President, SIAM Student Chapter 2022 – 2025
Astana IT University

- Organized a student meeting with representatives from **Boston University** and **Huawei**.
- Mentored **400+ first-year students** in Linear Algebra through peer-led academic support sessions.
- Helped organize seminars, workshops, and academic activities in applied mathematics and computer science.